2024 Second International Conference on Networks, Multimedia and Information Technology (NMITCON) | 979-8-3503-7289-2/24/$31.00 ©2024 IEEE | DOI: 10.1109/NMITCON62075.2024.10698932

Elevating Intrusion Detection Precision with Multi-Classification Algorithm Analysis

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***Abstract*— This paper examines the use of machine learning methods to detect network security threats. By analysing network traffic data and using a range of machine learning techniques. This paper focuses on improving IDS performance with ML techniques, utilizing the Kyoto20151207 dataset as a benchmark. It employs fifteen machine learning algorithms over a wide range of techniques to identify network breaches. This study aims to identify the most effective approach for intrusion detection tasks. Using comprehensive testing and assessment, several algorithm-performance couples are ranked according to correctness and accuracy. The study's goal is to give insight on the efficacy of various strategies, allowing practitioners to choose the best approaches for intrusion detection tasks. Furthermore, the accuracies of these methods are compared to those reported in other works, providing insight into the overall performance context. Finally, the study seeks to help practitioners make educated judgments when selecting relevant categorization algorithms for similar scenarios in IDS deployment. The key findings of our research are we implemented 15 different machine learning algorithms and identified the best accuracy in CatBoostClassifier algorithm as 99.738%.**

This paper's source code may be seen at the following website https://github.com/rashmika357/ideal-palm- tree

Index Terms*— *Intrusion detection, security, accuracy, algorithms

I. Introduction

The Intrusion Detection System (IDS) is a key component for detecting and mitigating potential security threats. IDS, which uses both hardware and software implementations, aids in the detection of hostile actions within networked settings, making it a cornerstone of security architecture. The major algorithms are K-nearest neighbor, CatBoost, Support Vector Machine, AdaBoost, Naive Bayes, and Linear Discriminant Analysis (LDA), Logistic Regression (LG), Light GBM (LGBM), Multi- Layer Perceptron (MLP), Quadratic Discriminant Analysis, Decision tree Classifier and Random Forest algorithm.

In today's interconnected world, the expansion of the Internet and widespread usage of computer networks has created new potential for communication, collaboration, and invention. However, these developments come with major cybersecurity dangers, as criminal actors attempt to exploit weaknesses in network systems for personal gain or evil intent Ensuring the integrity and availability of network resources and safeguarding sensitive data are top priorities for individuals, organizations, and governments alike. To address these cybersecurity concerns, strong and effective intrusion detection systems (IDS) that can promptly identify and neutralize such threats are necessary.

IDS is essential for spotting and stopping malicious activity or unauthorized access on computer networks. Systems for Numerous applications make use of management control and data acquisition, or SCADA. The application of these methods may provide security risks. The Random Forest method produced the best accuracy in this case [2]. They utilized GridSearchCV to find fraudulent accounts using machine learning methods. This technique is used to search the best parameters from given parameters [3]. CDC-IDS2018 Dataset is used in ML- based IDS for UAVs. In this paper [4] random forest has highest accuracy. In this effort, machine learning (ML) algorithms have shown to be helpful instruments, enabling the analysis of vast volumes of network traffic data and the detection of trends that point to questionable or malevolent behavior.

The Kyoto 20151207 dataset is an invaluable resource for evaluating the performance of IDS, as it provides complete data on network traffic that includes both normal operations and varied attacks. By pre-processing the dataset, storing categorical characteristics and fixing missing values, as well as using feature scaling to normalize the data and make sure that every feature contributes equally to the classification, classification algorithms can perform better. Before executing the code we eliminate the unnecessary input variables from the dataset Kyoto20151207. In this paper we enhanced the performance of machine learning classification algorithms by eliminating superfluous and insignificant variables from the dataset.

To compare the accuracy of different classification algorithms, identify the best method for enhancing intrusion

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detection system performance, and ascertain how effectively and precisely detect and categorize networking problems or concerns, several techniques are used. The study uses the Kyoto20151207 dataset. Comparing and assessing the results of many algorithms might help determine the best pairing for Intrusion detection in real-world network situations.

A. Major contribution

1. Based on the accuracy of each classification method, this research endeavor proposes a higher level of accuracy.
2. Various machine learning classification techniques have been applied to obtain findings with improved accuracy.
3. Researchers now have a standard method for evaluating intrusion detection accuracy thanks to this creative approach.
4. A unique intrusion detection technique was used to show the suspicious flow detection analysis approach.
5. Experiments with intrusion detection were conducted on the benchmark dataset, kyoto20151207. The outcomes demonstrate that the machine learning model that was recommended did better on the dataset.

The next parts of this paper are arranged according to the following structure. An overview of previous studies and a study of the literature on intrusion detection in network data are given in Section 2 of the article. The has an entirely new network traffic intrusion detection architecture, algorithm, and flow chart. In Section 3, the entire job has been clarified. In Section 4, this research study results the proposed accurate model are shown together with the accuracy comparisons with other works. Section 5 concludes the research.

II. Literature Survey

This literature survey has different paper accuracies of different algorithms and compare them. In the actual world, a variety of applications are operated on computer networks, and network security is unbreakable. It is essential to safeguard network systems from possible dangers in the linked world of today. As a result, intrusion detection systems (IDS) are being used widely.

For SCADA-based power systems, the Machine Learning-based IDS design framework composed of precise dataset preprocessing, balanced data augmentation, automated feature selection, and stringent training and testing protocols. We evaluated our proposed architecture for a power system utilizing SCADA and using a publicly accessible ORNL dataset.

The research paper [3] aims to identify and prevent bogus IDs and pages. Instagram's professional profiles are frequently used to spread bogus news and information

The main objective is to locate fake accounts and pages. This paper investigates the various IDs on Instagram. They employed ML methodologies including LR, NB, SVM, DT, and RF to ascertain the validity of an ID.

This study work's primary goal is to replicate an Intrusion Detection System (IDS) model that has previously been trained using several machine algorithms in a testbed environment, then evaluate each model's performance.

TABLE I ANALYTICAL COMPARISON WITH EXISTING METHODS.

Sl no	Authors	Title	Algorithm	Accuracy
1	Vikas et al	Comparative Analysis of Intrusion Detection Mechanism for WSN Using ML	For Decision Tree Classifier	98.02%
2	Marzia Zaman et al	Validation of a Machine Learning-Based IDS Design Framework Using ORNL Datasets for Power System With SCADA	They used 3 classification algorithms, cross-validation results with single dataset. In this random forest has highest accuracy.	97.34%
3	Subham Das et al	An Efficient approach to detect fraud Instagram accounts using supervised ml algorithms	In this paper, they used 5 classification Algorithms and they used GridsearchCV.. The Random Forest algorithm has highest accuracy before and after using GridSearchCV	98%
4	Ghufran Ahmed et al	Simulating ML-Based Intrusion Detection System for Unmanned Aerial Vehicles (UAVs) using COOJA Simulator	In this paper three ml algorithms are used. And the Random Forest algorithm has highest accuracy	98.11%

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III. Proposed Method

The dataset which is used has 191157 entries and 12 unique features. We investigate different test and train set sizes. Numerous attack types in the realm of cybersecurity provide major risks to people, companies, and organizations worldwide. Online attacks, user-to-root (U2R), DDoS (distributed denial of service), and DoS (denial of service) are common forms of assaults.

DDoS assaults include numerous hacked systems working together to flood a target system or network with more amount of traffic or requests. This volume of traffic substantially affects the target's normal operation, making it inaccessible to genuine users.

An effort by an unauthorized user to obtain elevated access to a system is known as a U2R (User-to-Root) assault. Their intention is to elevate their rights from those of a regular user to that of a root user, or system administrator. An attacker with the root authorisation can read confidential information, alter system configurations, and set up hidden entry points for additional assaults since they have total control over the system.

The packets are carefully examined to establish the classes and properties for each flow. In order to offer categorical data, we eliminated a large number of superfluous characteristics from the system model after adding the Kyoto20151207 dataset. In the following stage, data will split up into train and test sets.

After loading dataset into csv file data is split into feature variables (X) and target variable (y). Missing values are handled and datasets are splitted as train set and test set during data processing. By ensuring that the data from Training and Testing flows consistently across each column. Feature scaling is performed using Standard Scaler to standardize the feature variables to have mean 0 and variance 1. This helps in improving the performance of certain machine learning algorithms. Now we employ a particular model and this trained model is used to make predictions on test data. Results for accuracy are derived using algorithm.

Dataset: *https://www.takakura.com/Kyoto_data/new_data201704/*

Algorithm 1: Machine learning Algorithms

1. *Import Kyoto20151207 dataset and all libraries.*
2. *In data preprocessing, split data into train and test data.*
3. *Perform feature scaling on features using standard scaler*
4. *Apply the appropriate machine learning model.*
5. *Evaluate the model.*
6. *Calculate confusion matrix and accuracy*
7. *Print results*
8. *End*

.

Flow Chart:

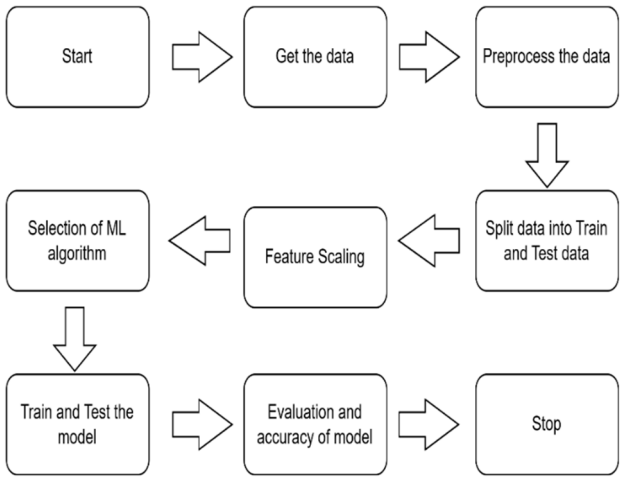

Fig. 1 Flow diagram of proposed work.

The machine learning algorithm's workflow is depicted in the Fig 1. Prior to analysis, the data is cleaned up and given the appropriate treatment by eliminating duplicates and managing missing values, among other preparation procedures. Afterwards, training and testing sets are created from the dataset.

To detect patterns and provide results, the machine learning model is taught using training data. The model's performance is then improved by using machine learning technique and evaluated using the testing dataset. After evaluation results are predicted for specific machine learning algorithms.

The following formulas are the mathematical expression to calculate the accuracy in our work.

$$Accurcay = 100\% - Rate\ error$$

$$Error\ rate = \frac{[Observedvalue - Actualvalue]}{Actualvalue} \times 100$$

IV. Results and Discussion

The accuracy of machine learning techniques on the kyoto20151207 dataset is the outcome of this paper. Ratios are used to split the test and train data 60:40, 70:30, and 80:20 are the ratios. The following graphs show the minimal qualities that were found to cut down on calculation time. It also shows how comparison analysis was used to carry out the most precise feature selection processes for the Kyoto20151207 database

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The Ada Boost Classifier is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using Ada Boost machine learning algorithm accuracy in Fig 2.

In this bar chart accuracy was recorded as 82.18% when training data split up is 70 and testing data split up is 30.

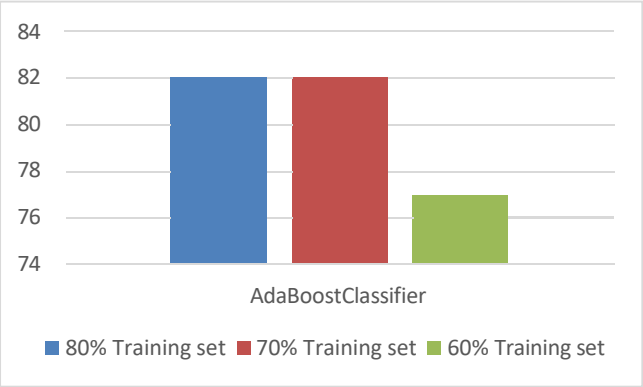

Fig. 2 Kyoto20151207 dataset using Ada Boost Classifier

The Cat Boost Classifier is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using Cat Boost machine learning algorithm accuracy in Fig 3.

In this bar chart accuracy was recorded as 99.7% when training data split up is 80 and testing data split up is 20.

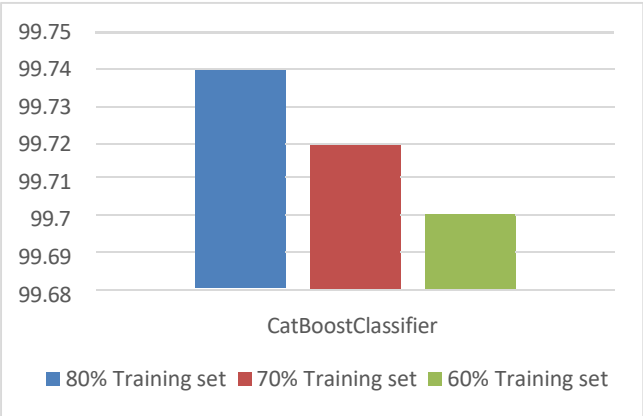

Fig. 3 Kyoto20151207 dataset using Cat Boost Classifier

The KNN Classifier is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using KNN machine learning algorithm accuracy in Fig 4.

In this bar chart accuracy was recorded as 98.18% when training data split up is 80 and testing data split up is 20.

Fig. 4 Kyoto20151207 dataset using KNN

The LDA Classifier is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using LDA machine learning algorithm accuracy in Fig 5. In This bar chart accuracy was recorded as 87.46% when training data split up is 70 and testing data split up is 30

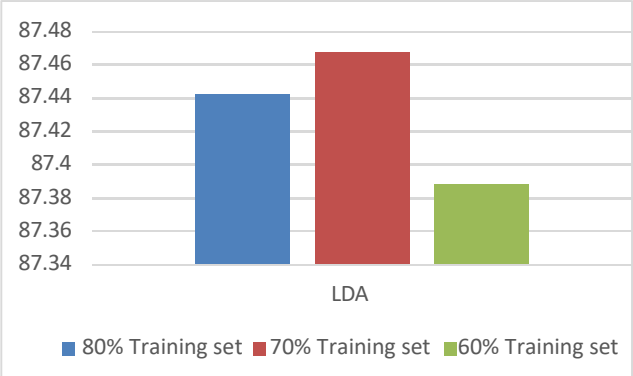

Fig. 5 Kyoto20151207 dataset using LDA

The LG Classifier is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using LG machine learning algorithm accuracy in Fig 6.

In this bar chart accuracy was recorded as 89.54% when training data split up is 70 and testing data split up is 30.

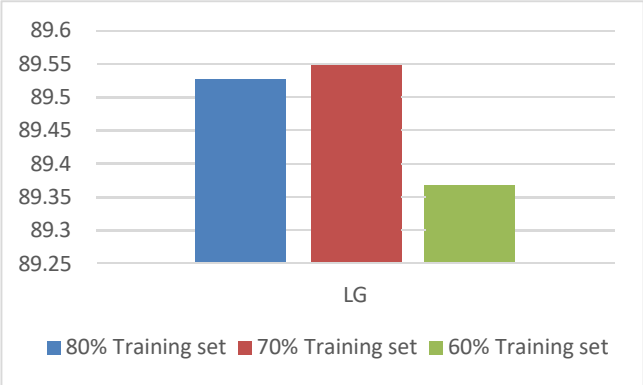

Fig. 6 Kyoto20151207 dataset using LG.

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The LGBM Classifier is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using LGBM machine learning algorithm accuracy in Fig 7.

In this bar chart accuracy was recorded as 99.73% when training data split up is 80 and testing data split up is 20.

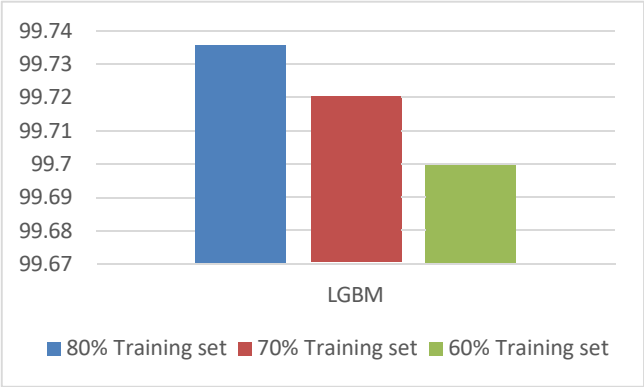

Fig. 7 Kyoto20151207 dataset using LGBM

The LSVM Classifier is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using LSVM machine learning algorithm accuracy in Fig 8.

In this bar chart accuracy was recorded as 92.00% when training data split up is 70 and testing data split up is 30.

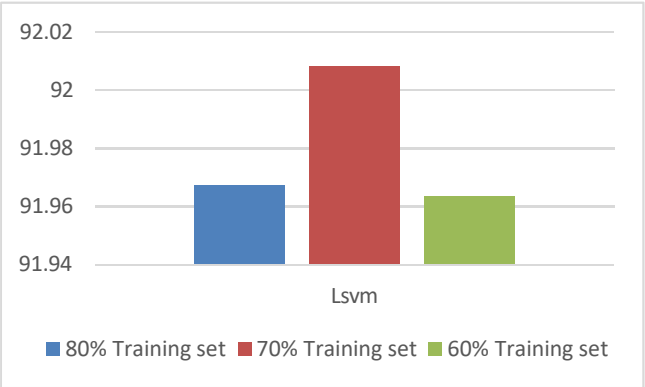

Fig. 8 Kyoto20151207 dataset using LSVM

The MLPC Classifier is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using MLPC machine learning algorithm accuracy in Fig 9.

In this bar chart accuracy was recorded as 96.29% when training data split up is 70 and testing data split up is 30, and 95.99% when training data split up is 80 and testing data split up is 20.

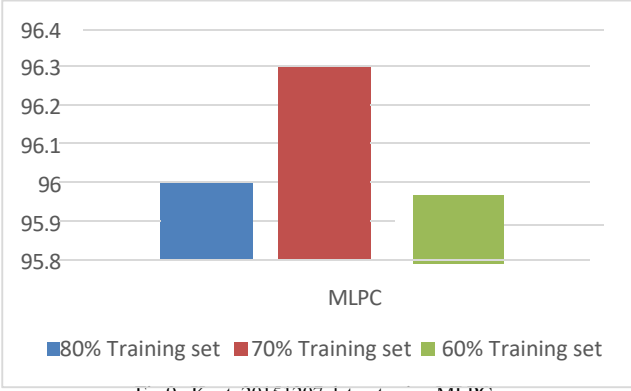

Fig 9 Kyoto20151207 dataset using MLPC

The Naïve Bayes Classifier is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using Naive Bayes machine learning algorithm accuracy in Fig 10.

In this bar chart accuracy was recorded as 86.32% when training data split up is 80 and testing data split up is 20.

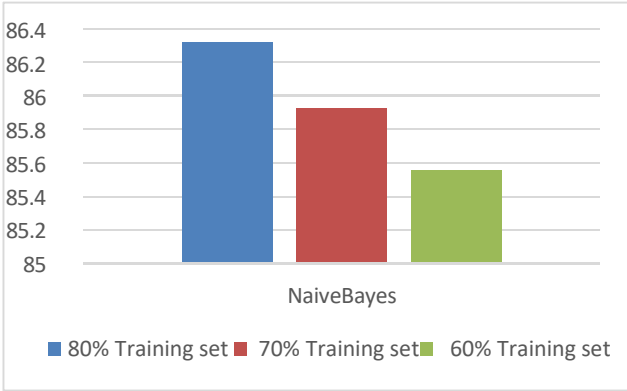

Fig. 10 Kyoto20151207 dataset using Naïve Bayes

This Quadratic Discriminant classifier is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using Quadratic Discriminant machine learning algorithm accuracy in Fig 11.

In this bar chart accuracy was recorded as 87.94% when training data split up is 70 and testing data split up is 30.

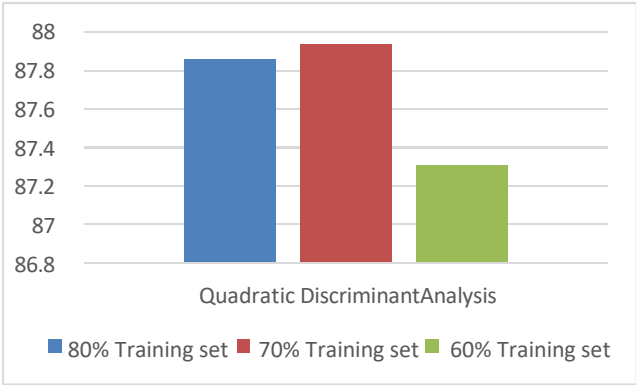

Fig. 11 Kyoto20151207 dataset using Quadratic Discriminant Analysis

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The XGB Classifier is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using XGB machine learning algorithm accuracy in Fig 12.

In this bar chart accuracy was recorded as 99.529% when training data split up is 60 and testing data split up is 40

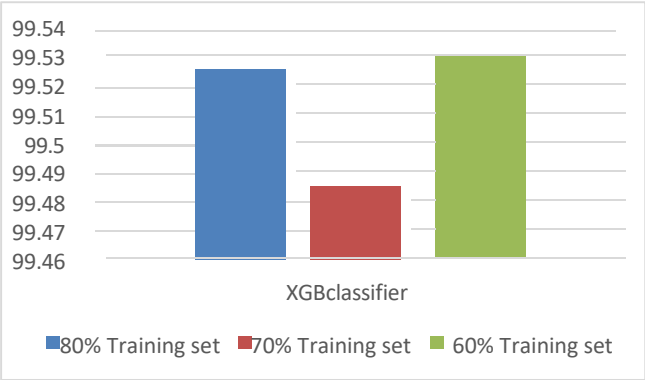


Fig. 12 Kyoto20151207 dataset using XGB Classifier

The Decision Tree Classifier is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using Decision Tree machine learning algorithm accuracy in Fig 13.

In this bar chart accuracy was recorded as 99.7% when training data split up is 80 and testing data split up is 20.

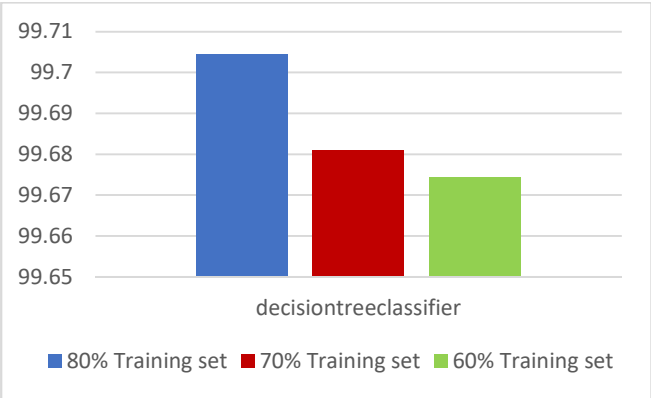
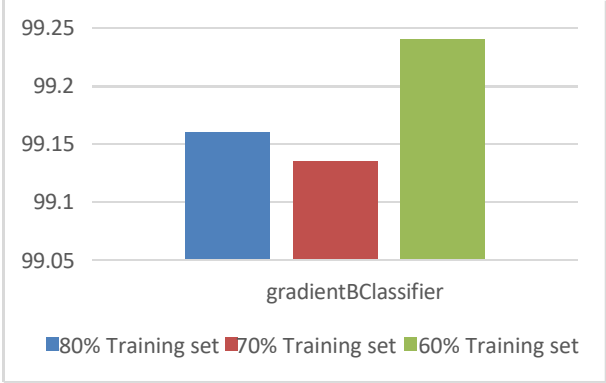


Fig. 13 Kyoto20151207 dataset using Decision Tree Classifier

The Gradient B Classifier is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using Gradient B machine learning algorithm accuracy in Fig 14.

In this bar chart accuracy was recorded as 99.2% when training data split up is 60 and testing data split up is 40.

Fig. 14 Kyoto20151207 dataset using gradientBClassifier



The KSVM is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using KSVM machine learning algorithm accuracy in Fig 15.

In this bar chart accuracy was recorded as 95.45% when training data split up is 60 and testing data split up is 40.

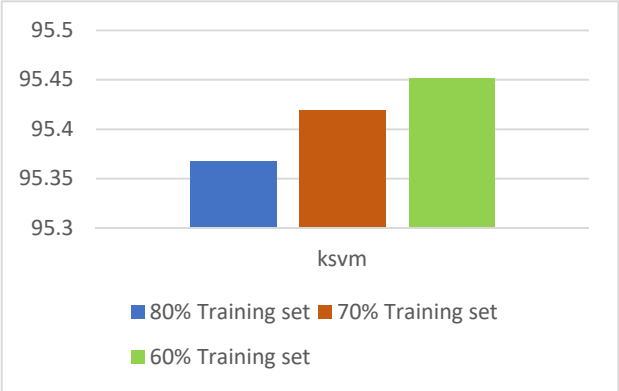


Fig. 15 Kyoto20151207 dataset using KSVM

The Random Forest Classifier is used to evaluate the kyoto20151207 dataset. The above bar chart gives information about split up of training data in 80,60,40 and also testing data 20,40,60 using Random Forest machine learning algorithm accuracy in Fig 16.

In this bar chart accuracy was recorded as 95.45% when training data split up is 60 and testing data split up is 40.

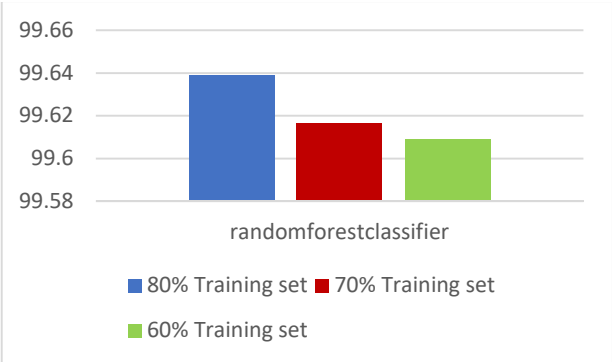


Fig. 16 Kyoto20151207 dataset using RandomForestClassifier

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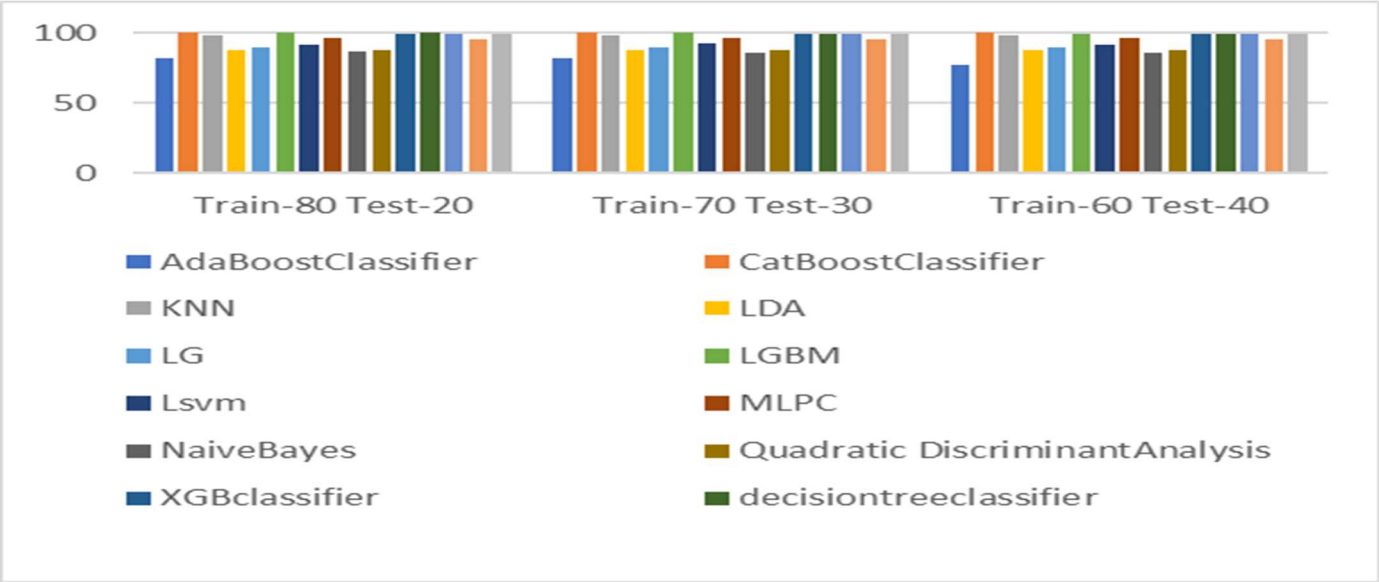

Fig. 17 Comparison of accuracies of all classification algorithms

The above bar graph that is Fig 16 shows the accuracies of different algorithms. Cat Boost Classifier is having highest accuracy for kyoto20151207 dataset that is 99.73% when train and test dataset ration is 80:20.

TABLE II COMPARISON OF ACCURACIES OF ALGORITHMS

Algorithms	Train-80 Test-20	Train-70 Test-30	Train-60 Test-40
AdaBoost	82.05	82.18	76.97
CatBoost	99.73	99.7175	99.7018
KNN	98.18	98.10	97.99
LDA	87.44	87.46	87.38
LG	89.52	89.54	89.36
LGBM	99.738	99.72	99.6
Lsvm	91.96	92.00	91.96
MLPC	95.99	96.296	95.97
NaiveBayes	86.32	85.92	85.55
QDA	87.86	87.94	87.30
XGBclassifier	99.52	99.48	99.52
decisiontree	99.70	99.68	99.67
gradientB	99.16	99.13	99.24
ksvm	95.36	95.41	95.45
randomforest	99.639	99.61	99.60

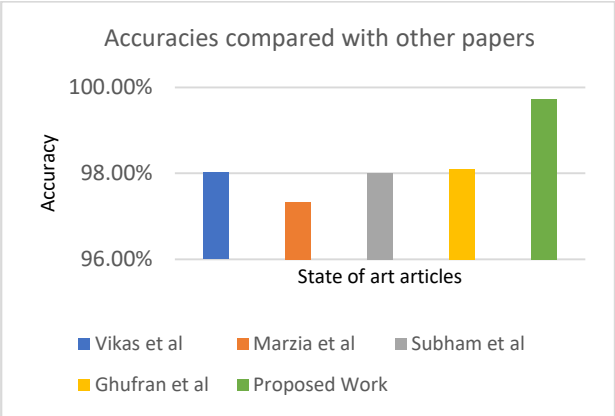

Fig. 18 Comparison of accuracies with the accuracies of previous papers

The Fig 18 bar graph shows accuracies of previous papers and the present paper used kyoto20151207 dataset and gained highest accuracy of 99.73% using cat boost classifier. The up- comping chart gives authors of the papers and it compared accuracies between different authors of the paper.

TABLE III COMPARISON OF ACCURACIES OF OTHER PAPERS

Authors	Algorithms	Accuracy
Vikas et al	Decision Tree	98.02%
Marzia et al	Random Forest	97.34%
Subham et al	Random Forest	98%
Ghufran et al	Random Forest	98.11%
Proposed Work	Cat Boost	99.73%

V. Conclusion

The Kyoto20151207 dataset is used to test the accuracy in terms of several algorithms in ML. The pertinent components were picked in order to raise performance throughout the training process. The dataset is preprocessed using a label encoder, a conventional scaler for feature scaling, and a data division into training data and testing data. Then the other classification methods are then applied. The accuracies are compared with previous papers. The best accuracy is obtained using the Cat Boost Classifier approach that is 99.73% for the dataset Kyoto20151207. The Decision tree approach has accuracy of 99.70% and random forest has accuracy of 99.63%. In the future work the research can be done using advanced machine learning algorithms with the kyoto20151207 dataset to analyze the accuracy with the proposed result.

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